Amir Beck, Chapter 1 - Exercise 1.2

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1.2. Prove that for any $\mathbf{x} \in \mathbb{R}^n$ one has

$$||\mathbf{x}||_\infty = \lim_{\mathbf{p}\to\infty} ||\mathbf{x}||_\mathbf{p}$$

Let $\mathbf{x} = (x_1, ..., x_n) \in \mathbb{R}^n$.

Let $M = \max_{i=1,...,n} |x_i| = ||\mathbf{x}||_\infty$ and let $k = \#\{1 \leq j \leq n \mid |x_j| = M\}$ the number of components of $\mathbf{x}$ that attained the max M in absolute value. Assume that $j_1, ..., j_k$ are the indices of the components that attained the maximum and $i_1, ..., i_{n-k}$ are the indices of the cordinates left.

Case 1: $(M \neq 0)$

$$\begin{aligned}
||\mathbf{x}||_p &:= (|x_1|^p + ... + |x_n|^p)^{1/p} \\
&= (|x_{j_1}|^p + ... + |x_{j_k}|^p + |x_{i_1}|^p + ... + |x_{i_{n-k}}|^p)^{1/p} \\
&= (kM^p + |x_{i_1}|^p + ... + |x_{i_{n-k}}|^p)^{1/p} \\
&= \left\{ M^p k \left[1 + \frac{1}{k}\left(\frac{|x_{i_1}|}{M}\right)^p + ... + \frac{1}{k}\left(\frac{|x_{i_{n-k}}|}{M}\right)^p \right] \right\}^{1/p} \\
&= M k^{1/p} \left[1 + \frac{1}{k}\left(\frac{|x_{i_1}|}{M}\right)^p + ... + \frac{1}{k}\left(\frac{|x_{i_{n-k}}|}{M}\right)^p \right]^{1/p} \\
&\underset{p\to\infty}{\longrightarrow} M \times 1 \\
&= M,
\end{aligned}$$

Note that since $M > |x_{i_1}|, ..., |x_{i_{n-k}}|$, we have that $\frac{|x_{i_1}|}{M}, ..., \frac{|x_{i_{n-k}}|}{M} < 1$ and thus

$$\left(\frac{|x_{i_1}|}{M}\right)^p, ..., \left(\frac{|x_{i_{n-k}}|}{M}\right)^p \underset{p\to\infty}{\longrightarrow} 0$$

Case 2: $(M = 0)$ Then $\mathbf{x} = (0, ..., 0)$. In this case, we have

$$\begin{aligned} ||\mathbf{x}||_p &= (|0|^p + ... + |0|^p)^{1/p} \\ &= 0 \\ &\underset{p\to\infty}{\longrightarrow} 0 = M \end{aligned}$$